

Phase D2 First-Half Synthesis and Second-Half Preview

Series D — Paper 3 Derivation Notes | Note D2.30 (#525)

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This work derives from and formalizes the inter-Self coordination architecture introduced in “Inter-Self Coordination via Shared Substrate / Full Aspect Integration” (Li, April 2026), the third paper in the CKS theory series following “Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems” (Li, April 2026) and “The Instinct/Reasoning Separation Outside the Model” (Li, April 2026).

Abstract

Note D2.30 is a synthesis note marking the completion of Phase D2’s first half (notes D2.01–D2.29, numbered #496–#524 in the series). It serves as a navigation artifact: mapping the 29 completed notes to the specific Paper 3 claims and Phase D1 sub-commitments they operationalize, identifying which D1 sub-commitments have complete operational prior-art coverage, which have partial coverage, and which (all five Claim 6 sub-commitments) have no coverage yet. It closes with a substantive preview of the Phase D2 second half (notes D2.31–D2.80), which will introduce population-scope operational aspects for Claim 6, multi-event governance frameworks, cross-organizational governance agreements, and governance health indicators — territory the first half did not enter.

1. Position of D2.30 in the Series

Phase D2 comprises approximately 80 operational-variant and decomposition notes for Paper 3. Each D2 note takes one or more Phase D1 foundational sub-commitments and specifies the operational governance they require in practice — the who, when, how, and under what authority conditions each sub-commitment is exercised. The first 29 notes (D2.01–D2.29) addressed the operational governance of Paper 3 Claims 1 through 5. Four of those notes (D2.18, D2.19, D2.24, D2.29) function as synthesis and diagnostic infrastructure rather than as single-sub-commitment coverage.

D2.30 is a synthesis note, not a sub-commitment note. Its purpose is navigational: any future session beginning Series D work, or any external reader assessing the prior-art record, should be able to read D2.30 and determine what operational territory D2.01–D2.29 has covered, at what

granularity, and what the second half must supply. The note maps each completed D2 note to its D1 parent sub-commitment and Paper 3 claim, provides an honest coverage assessment, and specifies what each category of second-half work will add.

2. First-Half Coverage by Claim Area

2.1 Claim 1 — Shared Substrate Lifecycle Governance

Claim 1 establishes the shared substrate as the architectural object of inter-Self coordination: temporarily constructed for a FAI event, perimeter-spanning across distinct Selves' home governance perimeters, with all six Paper 1 commitments holding within scope throughout its existence. Phase D1 decomposed Claim 1 into five sub-commitments: construction-as-temporary (D1.01), perimeter-spanning property (D1.02), all six Paper 1 commitments holding within scope (D1.03), dissolution (D1.04), and persistence policy as substrate content (D1.05). The seven D2 notes covering Claim 1 collectively formalize the complete operational lifecycle of the shared substrate:

- **D2.01** (construction governance requirements) operationalizes D1.01: the conditions under which a shared substrate may be initiated, the authority required across participating Selves' governance perimeters, and the minimum record the substrate must carry at construction.
- **D2.02** (dissolution governance requirements) and **D2.28** (emergency dissolution) together operationalize D1.04: the planned and unplanned pathways through which a shared substrate is terminated, with governance authority and evolution-feed hand-off requirements at each.
- **D2.03** (provenance chain) and **D2.17** (home perimeter integrity) operationalize D1.02: the operational requirements that ensure the perimeter-spanning property is maintained throughout a FAI event without compromise to either participating Self's home governance perimeter.
- **D2.04** (three rights at inter-Self scope) and **D2.05** (AI-as-mediator at inter-Self scope) operationalize D1.03: the operational form of the inspect-modify-override rights and the substrate-mediator role as they apply across the inter-Self perimeter.

2.2 Claim 2 — FAI Operation Governance

Claim 2 specifies Full Aspect Integration (FAI) as the canonical operation over the shared substrate, with the aspect as unit of exchange, full merge as architectural default, exchange bounded to reasoning-layer content, and three pattern variants and three persistence loci structuring the event space. Phase D1 decomposed Claim 2 into seven sub-commitments: aspect as exchange unit (D1.06), n-ary cardinality (D1.07), full-merge default (D1.08), three pattern variants (D1.09), exchange bounded to substrate content (D1.10), three persistence loci (D1.11), and FAI as inter-Self analog of Paper 2 mating (D1.12). Nine D2 notes cover Claim 2:

- **D2.06** (aspect contribution mechanics) operationalizes D1.06: the operational requirements for forming, structuring, and contributing an aspect as the unit of exchange at the inter-Self boundary.

- **D2.07** (conflict detection at merge) and **D2.08** (selective merge protocol) together operationalize D1.08: the operational governance for executing full merge as the default and for structured handling of detected conflicts that arise during merge.
- **D2.09** (provenance-carry-over merge requirements) and **D2.16** (exchange bounding verification) operationalize D1.10: the requirements ensuring merged content carries forward its provenance and that the instinct-layer exclusion is actively verified rather than assumed.
- **D2.20** (cooperation/competition variant) operationalizes D1.09: the operational governance for the competitive pattern variant, distinguishing it from cooperative and neutral variants in terms of authority requirements and conflict-handling defaults.
- **D2.21** (standing configurations) and **D2.22** (multi-Self governance) together operationalize D1.07 and D1.12: the governance of n-ary FAI events and the use of pre-authorized configuration to reduce per-event governance overhead for repeated interactions.
- **D2.27** (partial participation and withdrawal) operationalizes D1.11: the operational governance for the three persistence loci, including the conditions under which a participating Self may withdraw aspects or limit persistence at event close.

2.3 Claim 3 — Three-Tier Conflict-Handling Governance

Claim 3 develops the three-tier conflict-handling mechanism operating within the shared substrate: preserve as substrate-level state default, resolve via orchestration with rules as inspectable substrate content under explicit human authority, and escalate to humans across joint authority of participating Selves' governance. Phase D1 decomposed Claim 3 into four sub-commitments (D1.13–D1.16). Three D2 notes cover Claim 3:

- **D2.13** (inter-Self conflict registry structure) operationalizes D1.13: the operational requirements for the substrate structure that holds preserved conflicts as first-class addressable objects throughout the FAI event lifecycle.
- **D2.14** (escalation routing and response protocol) operationalizes D1.15: the operational governance for routing an escalation across participating Selves' governance authorities, including authority verification, response timelines, and record requirements.
- **D2.15** (orchestration rule authoring for conflict resolution) operationalizes D1.14: the governance for authoring, versioning, and updating the tier-selection and conflict-classification rules that determine which tier applies for each conflict class under each event configuration.

2.4 Claim 4 — Four-Locus Evolution-Feed Governance

Claim 4 specifies the four-locus evolution-feed mechanism at the FAI hand-off boundary: dissolution hand-off, per-mechanism feed structure, layer-routing rule with three-case explicitness, and asymmetric ingestion under per-perimeter governance. Phase D1 decomposed Claim 4 into five sub-commitments (D1.17–D1.21). Five D2 notes cover Claim 4:

- **D2.10** (action-feedback ingestion governance) and **D2.11** (DNA absorption governance protocol) together operationalize D1.18 and D1.21: the per-mechanism feed structure as it applies to action-feedback and DNA evolution respectively, with asymmetric ingestion

operationalized through per-perimeter authority over what each home substrate absorbs.

- **D2.23** (FAI and home governance temporal interleaving) and **D2.26** (FAI event chaining) together operationalize D1.17: the governance requirements at the dissolution boundary, including hand-off sequencing and special governance requirements when FAI events are chained.
- **D2.25** (provenance internalization governance) operationalizes D1.19: the requirements for maintaining the three-case layer-routing rule as content is internalized into home substrate, with provenance tags distinguishing DNA-layer, action-layer, and instinct-adjacent contributions.

2.5 Claim 5 — Configuration as Substrate Content Governance

Claim 5 commits configuration to the same substrate-content register with recursive applicability across all configurable FAI dimensions. Phase D1 decomposed Claim 5 into four sub-commitments (D1.22–D1.25). Two D2 notes specifically cover Claim 5:

- **D2.12** (FAI configuration authoring governance) operationalizes D1.22: the governance for authoring, versioning, and exercising authority over the full set of configurable FAI dimensions as substrate content under joint human authority.
- **D2.21** (standing configurations, also counted under Claim 2) operationalizes D1.24: the use of pre-authorized recursive configurations as the operational form through which configuration-of-configuration becomes substrate content rather than informal practice.

2.6 Synthesis Notes

Four D2 notes function as synthesis and diagnostic infrastructure rather than as single-sub-commitment operationalizations:

- **D2.18** (complete FAI event governance record): a comprehensive enumeration of the governance record a single FAI event must carry from construction through dissolution, integrating requirements from Claims 1–5.
- **D2.19** (twelve FAI event anti-patterns): catalogs the failure modes that arise when one or more first-half operational requirements is violated or circumvented, organized by claim area.
- **D2.24** (FAI event operational test suite): a structured set of operational tests for the governance requirements formalized in D2.01–D2.23.
- **D2.29** (inter-Self governance trust calibration): addresses the cross-cutting operational requirement that participating Selves' governance authorities maintain calibrated trust as the baseline posture for inter-Self coordination.

3. Coverage Assessment

3.1 Sub-commitments with complete operational coverage

The following D1 sub-commitments have dedicated treatment from one or more D2 notes providing full operational prior-art coverage: D1.01 (D2.01); D1.02 (D2.03, D2.17); D1.03 (D2.04,

D2.05); D1.04 (D2.02, D2.28); D1.06 (D2.06); D1.07 (D2.22); D1.08 (D2.07, D2.08); D1.09 (D2.08, D2.20); D1.10 (D2.09, D2.16); D1.11 (D2.27); D1.13 (D2.13); D1.14 (D2.15); D1.15 (D2.14); D1.17 (D2.23, D2.26); D1.18 (D2.10, D2.11); D1.19 (D2.25); D1.21 (D2.10); D1.22 (D2.12); D1.24 (D2.21). Nineteen of the twenty-five D1 sub-commitments for Claims 1–5 have complete coverage.

3.2 Sub-commitments with partial operational coverage

Six D1 sub-commitments have been touched by D2 notes but lack dedicated operationalization of their full scope:

- **D1.05** (persistence policy as substrate content): addressed obliquely through D2.01 and D2.03. A dedicated note on governance requirements for authoring and versioning the persistence policy itself has not been written.
- **D1.12** (FAI as inter-Self analog of Paper 2 mating): D2.21 addresses one operational form of the analog. Full mapping of how the three Paper 2 mating variants operate at inter-Self scope awaits dedicated treatment.
- **D1.16** (cross-perimeter escalation as architectural commitment): D2.14 covers the routing and response protocol. The operational requirements for keeping cross-perimeter escalation architecturally binding rather than a deployment option have not been given dedicated treatment.
- **D1.20** (instinct-takes-no-FAI-input commitment): touched through D2.10 and D2.25. The governance requirements for actively enforcing the exclusion as a detectable substrate property — not a convention — have not been given a dedicated note.
- **D1.23** (configuration-of-configuration also substrate content): touched through D2.12 and D2.21. The second recursive level of governance has not received the depth D1.22 received.
- **D1.25** (recursion bottoming at human-authored governance authority): implicit throughout Claim 5 notes but not given a dedicated operationalization specifying where the recursion terminates and what evidence the substrate must carry.

3.3 Sub-commitments with no operational coverage yet

All five Claim 6 sub-commitments have no Phase D2 coverage at the close of the first half:

- **D1.26** (joint authority across population-level perimeters): no D2 note yet.
- **D1.27** (accumulated FAI events as evolution mechanism): no D2 note yet.
- **D1.28** (six-rung substrate-mediator ladder): no D2 note yet.
- **D1.29** (calibrated-humility register): no D2 note yet.
- **D1.30** (biology-resonance positioning at population scope): no D2 note yet.

The gap for Claim 6 is structural, not incidental. The first half focused on the operational governance of a single FAI event and its immediate lifecycle. Claim 6 operates at the population register — accumulated events, many participating Selves, population-scale dynamics — a register that requires a distinct operational framing the first half did not enter.

4. Phase D2 Second-Half Preview (D2.31–D2.80)

4.1 Claim 6 Population-Scope Operational Aspects

The five Claim 6 sub-commitments (D1.26–D1.30) require operational treatment at the population register:

- **Joint authority across population-level perimeters** (D1.26): how governance authority is structured and exercised when accumulated FAI events operate across many Selves' governance perimeters simultaneously, at a scale that cannot be reduced to any single bilateral event's joint-authority model.
- **Accumulated FAI events as evolution mechanism** (D1.27): the operational governance of the evolution signal that emerges from many FAI events over time, including how it is recorded, attributed, and made available to participating Selves' home evolution mechanisms without requiring a centralized coordinator.
- **Six-rung substrate-mediator ladder** (D1.28): the operational commitments at rungs 5 and 6 — inter-Self mediation and population-scale coordination — including what the substrate must contain at each rung to satisfy the governance commitments the ladder names.
- **Calibrated-humility register** (D1.29): the operational requirements for maintaining the architectural honesty posture as population-scale deployments generate empirical observations, and how the governance record carries that evolution.
- **Biology-resonance positioning at population scope** (D1.30): the operational requirements for keeping the biological analogy bounded to its architectural load-bearing function without importing unjustified population-dynamics claims.

4.2 Multi-Event Governance

FAI governance at scale involves portfolios of simultaneous and sequential events rather than isolated single events. The second half will formalize: FAI event portfolio governance (managing many parallel events under a shared framework, including capacity constraints and governance-overhead management); cross-event provenance continuity (maintaining coherent provenance when a Self participates in multiple concurrent or successive events); and governance capacity and saturation (operational requirements that prevent governance capacity limits from becoming silent violations of the architectural commitments).

4.3 Cross-Organizational Governance Agreements

Practical inter-Self coordination requires formal agreements between organizations before any FAI event can be conducted under the architectural commitments. The second half will formalize: pre-event governance agreement structure (the operational requirements for agreements covering joint authority, conflict-handling tier selection, evolution-feed ingestion governance, and configuration authority); governance agreement versioning and amendment (how agreements evolve as organizations' governance structures change and how in-flight FAI events are affected); and default governance postures in the absence of complete agreement (the preserve-tier default

and minimal governance record requirements).

4.4 FAI Governance Health Indicators

Healthy operational governance includes early-signal indicators that governance capacity and posture are degrading before acute violations occur. The second half will formalize: governance health metrics (latency of escalation response, completeness of governance record, frequency of tier-selection disputes, provenance-gap rate); governance degradation patterns (early-warning signatures that precede the twelve anti-patterns formalized in D2.19); and governance recovery protocols (the operational requirements for restoring architectural compliance after a detected governance failure).

4.5 Remaining First-Half Partial Coverage

The second half will also supply dedicated treatment for the six partially covered D1 sub-commitments identified in §3.2 — D1.05, D1.12, D1.16, D1.20, D1.23, D1.25 — closing the coverage gaps and providing the operational prior-art depth those sub-commitments require.

5. What D2.01–D2.29 Establishes as a Body

As a body, the first 29 Phase D2 notes establish the operational prior art for a single, fully governed FAI event under the Paper 3 architectural commitments. A practitioner following D2.01 through D2.29 would know the governance requirements at every stage of a FAI event's lifecycle: what authority is required to construct the shared substrate; what operational requirements ensure the perimeter-spanning property is maintained; how aspects are formed, contributed, merged, and bounded; how conflicts are detected, preserved, routed through the three-tier mechanism, and resolved or escalated; how the evolution feed is structured, tagged, and ingested at each participating Self's home perimeter; how configuration is authored and made recursive; and what failure modes to watch for across all of these. The four synthesis notes (D2.18, D2.19, D2.24, D2.29) integrate these requirements into a unified governance record, a diagnostic anti-pattern catalogue, an operational test suite, and a trust calibration framework.

What this body does not yet provide is the operational governance for what happens at scale — when many Selves participate, when many events accumulate, when the population-level dynamics Claim 6 describes become operational requirements rather than architectural projections. That is the work of Phase D2's second half.

6. Conclusion

D2.30 marks a deliberate pause in Phase D2's progression. Nineteen of the twenty-five D1 sub-commitments for Claims 1–5 have complete operational coverage; six have partial coverage; and all five Claim 6 sub-commitments have no coverage yet. The partial and missing coverage is not a gap in the architecture — it is a gap in the prior-art record. Phase D2's second half will close the six partial-coverage gaps, supply the Claim 6 operational prior art that the extension claim

requires, and add the multi-event, cross-organizational, and governance-health operational depth that makes the prior art useful as a practical governance reference rather than as a collection of isolated single-event specifications. The note count at D2.30 is 29; the planned total for Phase D2 is approximately 80. The second half has approximately 50 notes to supply that remaining territory.

This defensive-publication note is part of the CKS derivation-note series. Notes are deposited on Zenodo as prior art to document the space of operational, architectural, and compositional commitments that follow from the CKS theory papers. The series does not introduce new theory; it maps what the published papers commit to, in forms useful for subsequent implementation, evaluation, and comparison work.